In vitro evaluation of genotoxicity of avocado (Persea americana) fruit and leaf extracts in human peripheral lymphocytes.

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Persea americana is much sought after both for the nutritional value of its fruit and the medicinal values of its various plant parts. A chromosomal aberration assay was undertaken to evaluate the potential genotoxicity of crude extracts from avocado fruits and leaves. Chromosomal aberrations were observed in cultured human peripheral lymphocytes exposed to separately increasing concentrations of 50% methanolic extracts of Persea americana fruit and leaves. The groups exposed to leaf and fruit extracts, respectively, showed a concentration-dependent increase in chromosomal aberrations as compared to that in a control group. The mean percentage total aberrant metaphases at 100 mg/kg, 200 mg/kg, and 300 mg/kg concentrations of leaf extract were found respectively to be 58 ± 7.05, 72 ± 6.41, and 78 ± 5.98, which were significantly higher ($p < 0.0001$ each) than that in the control group (6 ± 3.39). The mean percentage total aberrant metaphases at 100 mg/kg, 200 mg/kg, and 300 mg/kg concentrations of fruit extract were found to be 18 ± 5.49, 40 ± 10.00, and 52 ± 10.20, respectively, which were significantly higher ($p = 0.033$, $p < 0.0001$, and $p < 0.0001$, respectively) than that for control (6 ± 3.39). Acrocentric associations and premature centromeric separation were the two most common abnormalities observed in both the exposed groups. The group exposed to leaf extracts also showed a significant number of a variety of other structural aberrations, including breaks, fragments, dicentrics, terminal deletion, minutes, and Robertsonian translocations. The group exposed to leaf extract showed higher frequency of all types of aberrations at equal concentrations as compared to the group exposed to fruit extract.